

Are Small LLMs Ready for Coding Agents?

Can tight execution feedback turn structured small-model actions into accepted workflows?

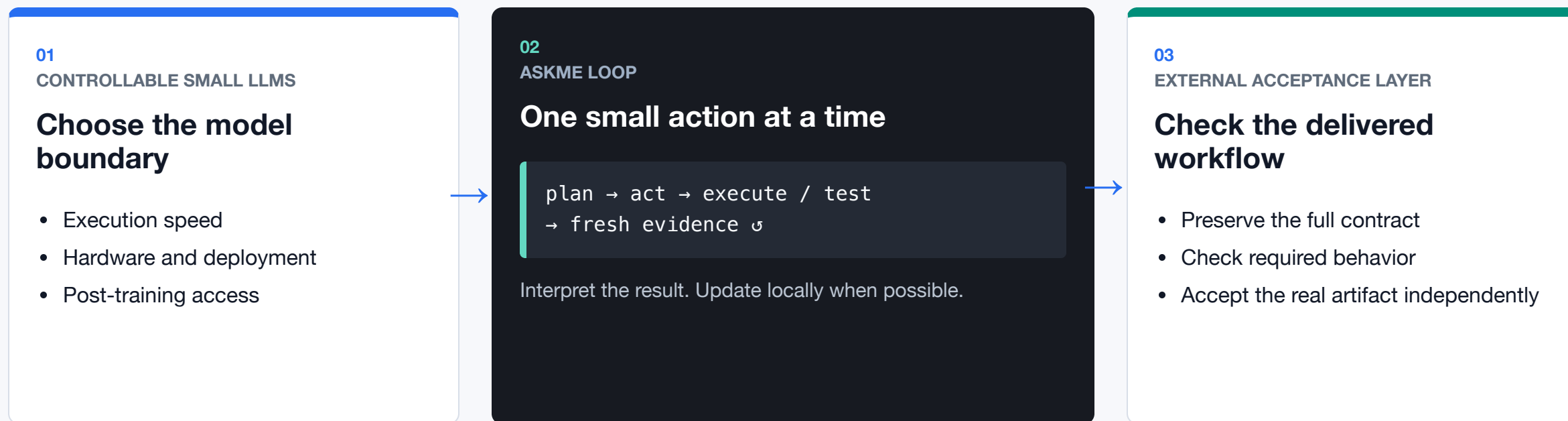
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[@den-run-ai](#) · github.com/den-run-ai/askme

AskMe gives a small model one structured move at a time

AskMe is an experimental coding-agent harness: explicit plan → one JSON action → execution evidence ☞



The bridge: fixed action vocabulary + fresh execution feedback + external workflow acceptance.

Assumes scoped actions, informative feedback, and independently testable success.

A command passed. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c
build ./main
run ./main
observe REPLAN_OK
```

A different target was chosen

```
cc -o /tmp/test main.c && /tmp/test
```

The combined command exited 0

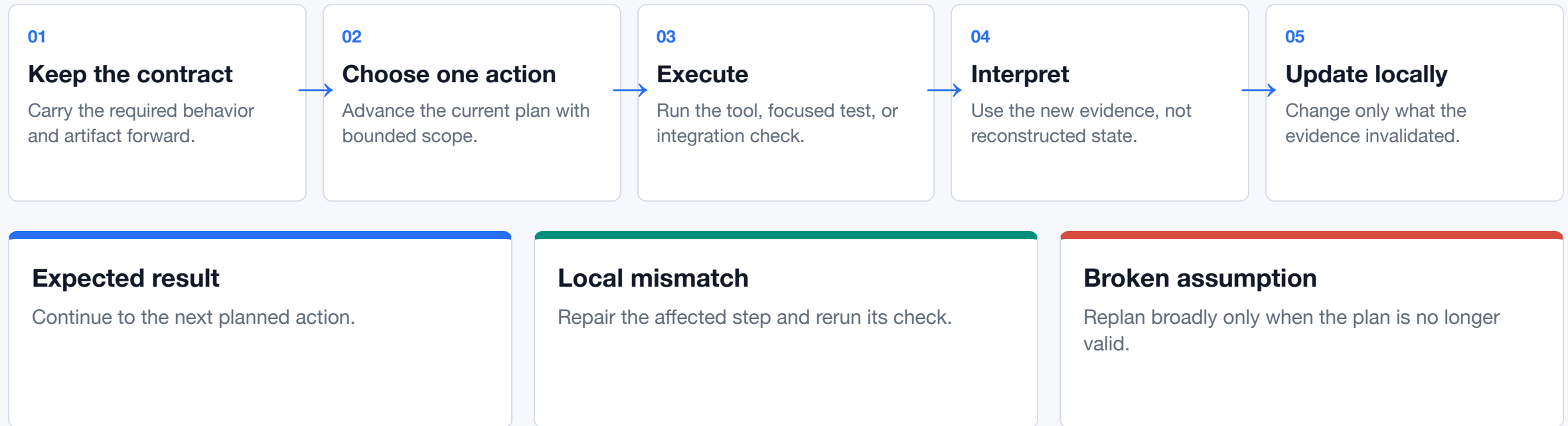
From the agent's side, the task looked done.

Agent reported completion

The independent acceptance test found no `./main`.

Exit-zero command + reported completion ≠ accepted artifact.

Hypothesis: update only what fresh evidence invalidates



Design goal: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn.

Acceptance caught the one bad deliverable

Four hosted models · two simple tasks · one run each

8 / 8

reported complete

7 / 8

artifacts accepted

The fastest build was the rejected one.
Completion and speed were not enough.

HOSTED VARIANT	ARTIFACT BUILD	SANITY REPAIR
Gemma 4 26B A4B	Accepted	Accepted
Gemma 4 31B	Accepted	Accepted
Qwen3.6-27B	Accepted	Accepted
Qwen3.6-35B-A3B	Wrong path	Accepted

Compatibility smoke, not a ranking: single runs on simple tasks — no model comparisons.

Both models build app features — but fail on testing

The same frozen task went from zero code changes to working partial features.

BEFORE · JULY

0 writes

Empty patch

The action interface blocked every edit. No code changed.

AFTER · AUG 1

App features built

→ **11 / 13** **7 / 13**

Gemma target tests Qwen target tests

Both patches applied — working partial features from both models.

WHY THEY STILL FAIL

They never test their work

→ Gemma rewrote code without running tests. Qwen stopped editing and went back to reading. Neither finished cleanly.

Bottom line: small models can build app features; testing and finishing the work is the next gap.

One task, one attempt per model — progress, not a benchmark score.

Promising for bounded loops. Feature readiness is still open.

Observed

Simple tasks: 7 / 8 artifacts accepted. Feature task: both models built working partial features — Gemma 11/13, Qwen 7/13 target tests — but neither tested or finished its work.

Supported

Harness design changed the outcome: the same task moved from empty patches to applied, partially working code.

Still open

Testing and clean completion. Reliability beyond one task. Model-to-model comparisons and local performance.

Evaluate the model, harness, and task as one system.

Current evidence supports boundary diagnosis—not a general readiness verdict.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data

A small model's workload depends on the harness

Three technical boundaries — not a leaderboard.

	AskMe	pi	OpenHands
ACTION SURFACE	6 fixed JSON actions; exactly one action per turn.	4 default tools; extensions can add or replace tools.	Typed, extensible <code>Action</code> → <code>Observation</code> tools.
STATE + CONTROL	Explicit plan, curated slim state, bounded local or full replanning.	Model-led session tree with branching and lossy compaction; no built-in plan mode.	Conversation state + append-only event log; optional persistence and configurable condenser.
COMPLETION BOUNDARY	<code>done</code> → conditional fail-open validation; held-out acceptance external.	Loop ends when tool calls stop; checks come from the workflow or extensions.	<code>finish</code> signals completion; benchmark evaluation remains a separate harness.

Trade-off, not ranking: AskMe spends more structure to reduce each turn's decision burden; pi keeps a minimal model-led core; OpenHands supplies a richer lifecycle runtime. All still need independent behavioral acceptance.